

Data Cleaning: Eurovision

Max Glonek

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Rule 1: Look at the Data

Rule 2: Is There a Package to Deal With That

```
songs <- jsonlite::fromJSON(glue::glue("./data/eurovision-lyrics.json"))
```

Rule 3: Break it Into Pieces

```
songs[[1]]
```

```
## $'#'  
## [1] "1"  
##  
## $Country  
## [1] "Netherlands"  
##  
## $#.1'  
## [1] "1"  
##  
## $Artist  
## [1] "Jetty Paerl"  
##  
## $Song  
## [1] "De vogels van Holland"  
##  
## $Language  
## [1] "Dutch"  
##  
## $Pl.  
## [1] "-"  
##  
## $Sc.  
## [1] "-"  
##  
## $Eurovision_Number  
## [1] 1  
##  
## $Year  
## [1] "1956"  
##  
## $Host_Country  
## [1] "Switzerland"  
##  
## $Host_City  
## [1] "Lugano"  
##  
## $Lyrics  
## [1] "De vogels van Holland zijn zo muzikaal\nZe leren in hun prille jeugd al tierelieren\nDe merel, c  
##  
## $'Lyrics translation'  
## [1] "The birds of Holland are so musical\nThey already learn to twitter in their early youth\nThe bl
```

```

parse_song <- function(song){
  country <- song$Country
  artist <- song$Artist
  title <- song$Song
  year <- song$Year
  lyrics <- song$Lyrics
  translation <- song$`Lyrics translation`
  place <- song$Pl.
  tibble(
    country, artist, title, year, lyrics, translation, place
  )
}
parse_song(songs[[1]])

```

```

## # A tibble: 1 x 7
##   country      artist      title      year lyrics  translation place
##   <chr>      <chr>      <chr>      <chr> <chr>   <chr>      <chr>
## 1 Netherlands Jetty Paerl De vogels van Holland 1956 "De vog~ "The birds~ -

```

Rule 4: Get it Into a Tibble

```

songs <-
  songs %>%
  map_df(parse_song, .id = "ID")
songs

```

```

## # A tibble: 1,644 x 8
##   ID      country      artist      title year lyrics  translation place
##   <chr> <chr>      <chr>      <chr> <chr> <chr>   <chr>      <chr>
## 1 0      Netherlands Jetty Paerl De v~ 1956 "De v~ "The birds~ -
## 2 1      Switzerland Lys Assia  Das ~ 1956 "Das ~ "The old c~ -
## 3 2      Belgium      Fud Leclerc Mess~ 1956 "Mess~ "Ye drowne~ -
## 4 3      Germany (West) Walter Andreas Sch~ Im W~ 1956 "Es g~ "There is ~ -
## 5 4      France        Mathé Altéry Le t~ 1956 "Chan~ "Sing, car~ -
## 6 5      Luxembourg    Michèle Arnaud Ne c~ 1956 "Si o~ "If they t~ -
## 7 6      Italy          Franca Raimondi Apri~ 1956 "La p~ "The first~ -
## 8 7      Netherlands Corry Brokken Voor~ 1956 "Voor~ "Over for ~ -
## 9 8      Switzerland Lys Assia (2) Refr~ 1956 "[Int~ "Chorus of~ 1
## 10 9     Belgium      Mony Marc  Le p~ 1956 "Les ~ "The bells~ -
## # i 1,634 more rows

```

Rule 5: Look at Each Column

We need to make sure all of our data makes sense in context. You can find the source data at this link: <https://www.kaggle.com/minitree/eurovision-song-lyrics>

ID

This is an ID number assigned to each song. We should check that the song IDs are all unique.

```
songs %>% count(ID) %>% filter(n > 1)
```

```
## # A tibble: 0 x 2
## # i 2 variables: ID <chr>, n <int>
```

They are unique, but they're also not useful, so delete them.

```
songs <-
  songs %>%
    select(-ID)
songs
```

```
## # A tibble: 1,644 x 7
##   country      artist      title    year lyrics translation place
##   <chr>        <chr>      <chr>    <chr> <chr> <chr>      <chr>
## 1 Netherlands Jetty Paerl    De voge~ 1956 "De v~ "The birds~ -
## 2 Switzerland Lys Assia     Das alt~ 1956 "Das ~ "The old c~ -
## 3 Belgium      Fud Leclerc   Messieu~ 1956 "Mess~ "Ye drowne~ -
## 4 Germany (West) Walter Andreas Schwarz Im Wart~ 1956 "Es g~ "There is ~ -
## 5 France        Mathé Altéry  Le temp~ 1956 "Chan~ "Sing, car~ -
## 6 Luxembourg    Michèle Arnaud Ne croi~ 1956 "Si o~ "If they t~ -
## 7 Italy          Franca Raimondi Aprite ~ 1956 "La p~ "The first~ -
## 8 Netherlands Corry Brokken Voorgoe~ 1956 "Voor~ "Over for ~ -
## 9 Switzerland Lys Assia (2) Refrain  1956 "[Int~ "Chorus of~ 1
## 10 Belgium      Mony Marc    Le plus~ 1956 "Les ~ "The bells~ -
## # i 1,634 more rows
```

Artist

Are there any artists who appear more than once?

```
songs %>%
  count(artist) %>%
  arrange(-n)
```

```
## # A tibble: 1,621 x 2
##   artist      n
##   <chr>    <int>
## 1 Ana Soklič      2
## 2 Benny Cristo    2
## 3 Blas Cantó      2
## 4 Daði og Gagnamagnið 2
## 5 Destiny         2
## 6 Eden Alene       2
## 7 Efendi           2
## 8 Gjon's Tears     2
## 9 Go_A             2
## 10 Hooverphonic     2
## # i 1,611 more rows
```

```
songs %>%
  filter(str_detect(artist, "Jed"))
```

```
## # A tibble: 2 x 7
##   country artist      title      year lyrics      translation place
##   <chr>   <chr>      <chr>    <chr> <chr>      <chr>    <chr>
## 1 Ireland Jedward    Lipstick  2011 "You say you're on it b~ English    8
## 2 Ireland Jedward (2) Waterline 2012 "Floodgates, can't wait~ English    19
```

```
songs %>%
  filter(country == "Australia")
```

```
## # A tibble: 7 x 7
##   country artist      title      year lyrics      translation place
##   <chr>   <chr>      <chr>    <chr> <chr>      <chr>    <chr>
## 1 Australia Guy Sebastian Tonight Again 2015 "[Verse~ English    5
## 2 Australia Dami Im    Sound of Silence 2016 "[Verse~ English    2
## 3 Australia Isaiah    Don't Come Easy 2017 "[Verse~ English    9
## 4 Australia Jessica Mauboy We Got Love 2018 "[Verse~ English   20
## 5 Australia Kate Miller-Heidke Zero Gravity 2019 "[Verse~ English    9
## 6 Australia Montaigne  Don't Break Me 2020 "I don'~ English    -
## 7 Australia Montaigne  Technicolour 2021 "I wann~ English    -
```

Title

```
songs %>%
  count(title) %>%
  arrange(-n)
```

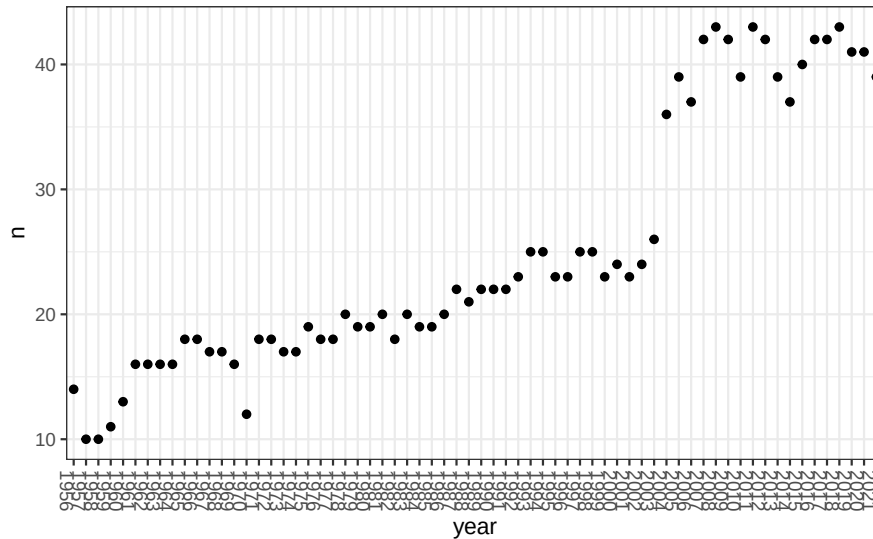
```
## # A tibble: 1,608 x 2
##   title      n
##   <chr>    <int>
## 1 Shine      4
## 2 Angel      3
## 3 Time       3
## 4 Amen       2
## 5 Attention  2
## 6 Casanova  2
## 7 Coming Home 2
## 8 Congratulations 2
## 9 Fairytale  2
## 10 Goodbye   2
## # i 1,598 more rows
```

Year

```
range(songs$year)
```

```
## [1] "1956" "2021"
```

```
songs %>%  
  count(year) %>%  
  ggplot(aes(year, n)) +  
  geom_point() +  
  theme(axis.text.x = element_text(angle = -90, hjust=0))
```



Lyrics

```
songs %>%  
  sample_n(5) %>%  
  pull(lyrics) %>%  
  cat(sep = "\n\n\n")
```

```
## [Strofa 1]  
## Essere o dover essere  
## Il dubbio amletico  
## Contemporaneo come l'uomo del Neolitico  
## Nella tua gabbia 2x3 mettiti comodo  
## Intellettuali nei caffè  
## Internettologi  
## Soci onorari al gruppo dei selfisti anonimi  
## L'intelligenza è démodé  
## Risposte facili  
## Dilemmi inutili  
##  
## [Bridge 1]  
## AAA cercasi (cerca sì)  
## Storie dal gran finale  
## Spèrasi (spera sì)  
## Comunque vada panta rei  
## And singing in the rain
```

```

##
## [Ritornello]
## Lezioni di Nirvana
## C'è il Buddha in fila indiana
## Per tutti un'ora d'aria, di gloria (alé)
## La folla grida un mantra
## L'evoluzione inciampa
## La scimmia nuda balla
## Occidentali's Karma
## Occidentali's Karma
## La scimmia nuda balla
## Occidentali's Karma
##
## [Strofa 2]
## Piovono gocce di Chanel
## Su corpi asettici
## Mettiti in salvo dall'odore dei tuoi simili
## Tutti tuttologi col web
## Coca dei popoli
## Oppio dei poveri
##
## [Bridge 2]
## AAA cercasi (cerca sì)
## Umanità virtuale
## Sex appeal (sex appeal)
## Comunque vada panta rei
## And singing in the rain
##
## [Ritornello]
## Lezioni di Nirvana
## C'è il Buddha in fila indiana
## Per tutti un'ora d'aria, di gloria
## La folla grida un mantra
## L'evoluzione inciampa
## La scimmia nuda balla
## Occidentali's Karma
## Occidentali's Karma
## La scimmia nuda balla
## Occidentali's Karma
##
## [Strofa 3]
## Quando la vita si distrae cadono gli uomini
## Occidentali's Karma
## Occidentali's Karma
## La scimmia si rialza
## Namasté, alé
##
## [Ritornello]
## Lezioni di Nirvana
## C'è il Buddha in fila indiana
## Per tutti un'ora d'aria, di gloria
## La folla grida un mantra
## L'evoluzione inciampa
## La scimmia nuda balla

```

Occidentali's Karma
 ## Occidentali's Karma
 ## La scimmia nuda balla
 ## Occidentali's Karma
 ##
 ##
 ## Tavo lūpos kvepia lietumi
 ## Nuodėmė paliesti jas, išsipildymo meldžiu
 ## Sapnas mano į tave įeis
 ## Likti tamsoje vienai neleis
 ##
 ## Mano neliesta styga
 ## Tu man neliesta
 ## Mano sapnas į tave
 ## Lieja svaigų vyną mūsų naktyje
 ##
 ## Jau ir visi nakties žiedai akis užmerkė
 ## Tik tavo toks gilus dangus
 ## Gėlės kaip ir tu be meilės verkia
 ## Ir sapnuoja tous pačius sapnus
 ##
 ## Mano neliesta styga
 ## Tu man neliesta
 ## Mano sapnas į tave
 ## Lieja svaigų vyną naktyje
 ##
 ## Mano neliesta styga
 ## Tu man neliesta
 ## Mano sapnas į tave
 ## Lieja svaigų vyną naktyje
 ##
 ##
 ## Don't try to deny
 ## I can see in your eyes
 ## All the love you'd like
 ## To hide away
 ##
 ## So don't cry, say goodbye
 ## And go on with your lies
 ## I know you are mine
 ## You'll come back someday
 ##
 ## Hasta la vista, baby
 ## Baby, so long
 ## Hasta la vista, baby
 ## Oh go, you are so...
 ##
 ## Hasta la vista, baby
 ## Baby, so long
 ## Hasta la vista, baby
 ## Oh go, you are so wrong
 ##
 ## Can't you see the signs?
 ## I am yours, you are mine


```

## You are sayin' goodbye
## I know you are
##
## Just try and rely
## To the voice of your heart
## Your tears in your eyes
## Are so bright as stars
##
## Hasta la vista, baby
## Baby, so long
## Hasta la vista, baby
## Oh go, you are so wrong
##
## You want to go
## You say goodbye
## You'd like to fly
## Up to the sky
##
## But I don't know (I don't know)
## So tell me why (Tell me why)
## You always cry (Cry)
## Just tell me why
##
## Hasta la vista...
## (Hasta la vista)
##
## Hasta la vista, baby
## Baby, so long
## Hasta la vista, baby
## Oh go, you are so...
##
## Hasta la vista, baby
## Baby, so long
## Hasta la vista, baby
## Oh go, you are so wrong
##
## Goodbye, you are so...
## Hasta la vista
## (Hasta la vista...)
##
##
## [Couplet 1]
## Quelque chose me manque, mais quoi? Je veux-ci je veux ça!
## Quand j'dors, j'fait des rêves en dollars, tous les jours, j'ai un nouveau costard
## Chez moi tout est neuf, tout est beau, le monde pleure derrière mes rideaux
## J'm'en fous, j'habite au dernier étage, j'connais même pas ma femme de ménage
## Y'a du cuir, dans ma voiture, l'odeur de mon parfum me rassure
## Je n'aime pas montrer mes émotions à la salle de musculation
## Je soulève quelques poids et haltères, mon corps est une machine de guerre
## J'ai tout ce qu'on rêverait d'avoir, j'ai peut-être tout c'est vrai mais moi j'voulais une moustache
##
## [Pré-refrain]
## Une moustache, une moustache
## I wanna have a moustache

```

```

## Une moustache, une moustache
## Quería un bigote
##
## [Refrain]
## Je veux ci je veux ça, y'en a jamais assez pour moi
## C'est comme-ci c'est comme ça, y'a toujours un truc que j'ai pas
## Je veux ci je veux ça, y'en a jamais assez pour moi
## C'est comme-ci c'est comme ça, y'a toujours un truc que j'ai pas
## Je veux ci je veux ça
## C'est comme-ci c'est comme ça
## Je veux ci je veux ça
## C'est comme-ci c'est comme ça, y'a toujours un truc que j'ai pas
##
## [Couplet 2]
## Alors j'en rajoute encore, des tonnes de choses derrière mes placards
## J'ai des amis quand même et même j'ai des amis qui m'aiment
## J'me dit j'ai tout pour plaire, j'ai tout pour, j'ai le monde à mes pieds (C'est fou)
## J'ai tout ce qu'on rêverait d'avoir, j'ai peut-être tout c'est vrai mais moi j'avais une moustache
##
## [Pré-refrain]
## Une moustache, une moustache
## I wanna have a moustache
## Une moustache, une moustache
## Quería un bigote
##
## [Refrain]
## Je veux ci je veux ça, y'en a jamais assez pour moi
## C'est comme-ci c'est comme ça, y'a toujours un truc que j'ai pas
## Je veux ci je veux ça, y'en a jamais assez pour moi
## C'est comme-ci c'est comme ça, y'a toujours un truc que j'ai pas
##
## [Pont]
## J'donnerais tout ce que j'ai contre une moustache
## Maintenant je sais, ce que je veux c'est une moustache
## J'pourrais tout laisser, tout donner pour une moustache
## J'ai peut-être tout c'est vrai mais pas ce que je voulais
##
## [Refrain]
## Je veux ci je veux ça, y'en a jamais assez pour moi
## C'est comme-ci c'est comme ça, y'a toujours un truc que j'ai pas
## Je veux ci je veux ça, y'en a jamais assez pour moi
## C'est comme-ci c'est comme ça, y'a toujours un truc que j'ai pas
##
## [Outro]
## Mais moi j'avais une moustache
##
##
## She smiled at me, said: "Would you like it sweet and low?"
## "I know a place where we can take it nice and slow"
## I know what you're thinking: don't stop, I need that sweet thing
## Telling you now this is real
## Shake it up 'eakerim, I know what you're feelin'
## Shake it up 'eakerim, I got what you're needin'
## Shake it up 'eakerim, tell me what you're thinkin'

```

```
## Shake it up ºekerim, you're my only sweet thing
## She said in a sexy lazy low key: "Can I make it right?"
## You put your trust on me, I'll take you through the night
## Don't keep me waiting, not sure you're understanding
## Just how much I crave for you
## Shake it up ºekerim, I know what you're feelin'
## Shake it up ºekerim, I got what you're needin'
## Shake it up ºekerim, tell me what you're thinkin'
## Shake it up ºekerim, you're my only sweet thing
## Lovey-dovey, lovey-dovey all the time
## I got lots of candy to make you mine
## Lovey-dovey, lovey-dovey all the time
## I got lots of candy for you
## Lovey-dovey, lovey-dovey all the time
## I got lots of candy to make you mine
## Lovey-dovey, lovey-dovey all the time
## I got lots of candy for you
## Shake it up ºekerim, I know what you're feelin'
## Shake it up ºekerim, I got what you're needin'
## Shake it up ºekerim, tell me what you're thinkin'
## Shake it up ºekerim, you're my only sweet thing
## (Shake it up ºekerim) Shake it up ºekerim
## Shake it up ºekerim, you're my only sweet thing
## Shake it up ºekerim, tell me what you're thinkin'
## Shake it up ºekerim, you're my only sweet thing
## Shake it up ºekerim
```

Translation

```
songs %>%
  sample_n(5) %>%
  pull(translation) %>%
  cat(sep = "\n\n\n")
```

```
## The revelry was unrestrained
## The hangover is endless
## Life is meaningless
## The void will get us all Hate will prevail
## Happiness comes to an end
## For it is an illusion
## A treacherous pipe dream All that I saw
## Tears ran down
## All that I gave
## Once gave
## I gave it all to you Multilateral delusions
## Unilateral punishments
## Gullible poor fellows
## The escape will end
## The emptiness will get us all Hate will prevail
## Europe will crumble
## A web of lies
## Will arise from the ashes
```

United as one All that I saw
Tears ran down
All that I gave
Once gave
I gave it all to you All that I saw
Tears ran down
All that I gave
Once gave
I gave it all to you Hate will prevail
Love will die
Hate will prevail
Happiness comes to an end
For it is an illusion
A treacherous pipe dream Hate will prevail

English

My journey was to the sun, far beyond any enclosed spaces,
Where everything is infinite, and all borders are erased.
I want to see the miracles and hear the words which create life
Being carried by a force, which only grows when I imagined my motives City of light, in a nameless c
Give me life, where everything is born again Then when everything changes and time doesn't exist any
That's when I'll see us together, and the journey is over.
That is the only thing we can remember City of light, in a nameless country
Give me life, where everything is born again
City of life, in a nameless country
Give me life, where everything is born again

When you're here
When you are gone
And though all the birds are singing
When you are gone
And though the flowers are blooming
I'm lonely
And all the people around me are behaving
As if nothing had happened
When you are gone
I don't hear the birds' song
When you are gone
The sun itself shines dimly
Whatever I do, everything is bland and lost
Then I miss you so much
But when you are here
And I breathe close to you
When you are close to me
And I look in the eyes
When you are here
And I take you in my arms
We are forgetting the world around us
When you are here
I see the tree blooming again

```
## When you are here
## I see the flowers blooming again
## When you are here
## I wish so much:
## Please never don't go again
## When you are gone
## I walk around without any aim
## Totally lost
## I feel hollowed and empty
## When you are gone
## I have no more plans
## Then I miss you so much
## But when you are here
## And I breathe close to you
## When you are close to me
## And I look in the eyes
## When you are here
## And I take you in my arms
## We are forgetting the world around us
## When you are here
## I see the tree blooming again
## When you are here
## I see the flowers blooming again
## When you are here
## I wish so much:
## Please never don't go again
##
##
## English
```

Place

```
songs %>% count(place)
```

```
## # A tibble: 27 x 2
##   place      n
##   <chr> <int>
## 1 -      309
## 2 1       68
## 3 10      66
## 4 11      58
## 5 12      60
## 6 13      70
## 7 14      59
## 8 15      53
## 9 16      56
## 10 17      48
## # i 17 more rows
```

```
length(unique(songs$year))
```

```
## [1] 66
```

```
songs %>%
  count(year, place) %>%
  filter(place == "1") %>%
  arrange(-n)
```

```
## # A tibble: 65 x 3
##   year place    n
##   <chr> <chr> <int>
## 1 1969 1         4
## 2 1956 1         1
## 3 1957 1         1
## 4 1958 1         1
## 5 1959 1         1
## 6 1960 1         1
## 7 1961 1         1
## 8 1962 1         1
## 9 1963 1         1
## 10 1964 1         1
## # i 55 more rows
```

According to Wikipedia, there were four winners in 1969:

https://en.wikipedia.org/wiki/Eurovision_Song_Contest

On only one occasion have multiple winners been declared in a single contest: in 1969, four countries finished the contest with an equal number of votes and due to the lack of a tie-break rule at the time, all four countries were declared winners.[7][157]

Rule 6: You May Need to Go Back

Our original parse function just grabbed `translation` as is. After looking at `translation`, we can see that this variable says `English` if the original song lyrics are in English. English.

Rule 7: Create New Columns?

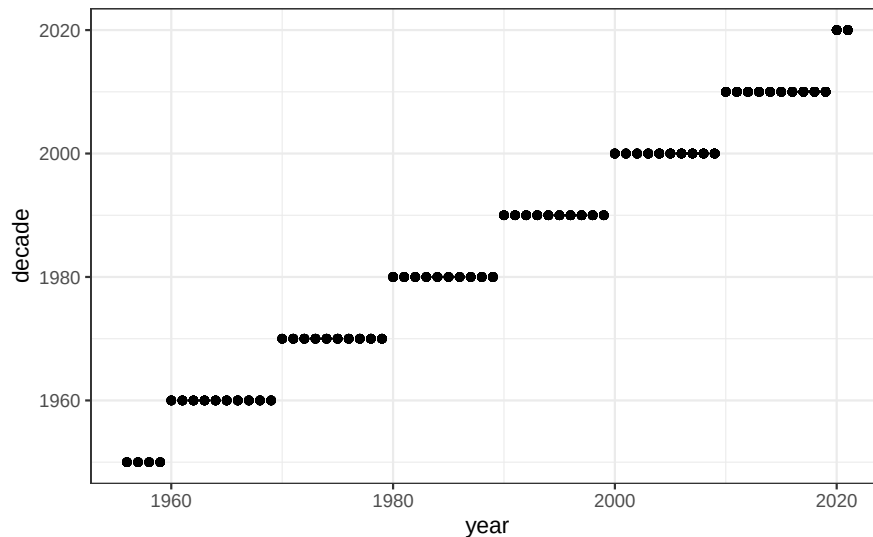
Decade

```
songs
```

```
## # A tibble: 1,644 x 7
##   country      artist      title    year lyrics translation place
##   <chr>      <chr>      <chr>    <chr> <chr> <chr>      <chr>
## 1 Netherlands Jetty Paerl  De voge~ 1956 "De v~ "The birds~ -
## 2 Switzerland Lys Assia   Das alt~ 1956 "Das ~ "The old c~ -
## 3 Belgium     Fud Leclerc Messieu~ 1956 "Mess~ "Ye drowne~ -
## 4 Germany (West) Walter Andreas Schwarz Im Wart~ 1956 "Es g~ "There is ~ -
## 5 France      Mathé Altéry Le temp~ 1956 "Chan~ "Sing, car~ -
## 6 Luxembourg  Michèle Arnaud Ne croi~ 1956 "Si o~ "If they t~ -
```

```
## 7 Italy          Franca Raimondi      Aprite ~ 1956 "La p~ "The first~ -
## 8 Netherlands   Corry Brokken        Voorgoe~ 1956 "Voor~ "Over for ~ -
## 9 Switzerland   Lys Assia (2)      Refrain  1956 "[Int~ "Chorus of~ 1
## 10 Belgium      Mony Marc          Le plus~ 1956 "Les ~ "The bells~ -
## # i 1,634 more rows
```

```
songs <-
  songs %>%
  mutate(
    year = parse_number(year),
    decade = year - year %% 10
  )
songs %>%
  ggplot(aes(year, decade)) +
  geom_point()
```



Lyrics

```
songs %>%
  count(translation) %>%
  arrange(-n)
```

```
## # A tibble: 946 x 2
##   translation                                     n
##   <chr>                                           <int>
## 1 "English"                                     694
## 2 ""                                              2
## 3 "Day is turning to evening, is gone, soon forgotten\nBut what we gave ~ 2
## 4 "God, god, god, god they don't call you\nCandles, candles, candles, ca~ 2
## 5 "Look into my eyes now,\nI'm not a loser, know it!\nI don't surrender~ 2
## 6 "Only champions!\nOnly champions! They fly high1 and jump far2.\nAll t~ 2
## 7 "\nPure and white, her skin is so inviting\nLike a new, unbeaten snow~ 1
## 8 "\"Come with me\", said my friend\nThe first silent winter snow is fal~ 1
```

```
## 9 "\"This is the Telephone Information Service, hello\\n\\n\"Yes, hello, t~ 1
## 10 "\"We are all moving around the old circle\\n1,\\nSinging in this sweet ~ 1
## # i 936 more rows
```

```
songs <-
  songs %>%
  mutate(
    english_lyrics = ifelse(
      translation == "English",
      lyrics, translation
    )
  )
```

Rule 7: Save the Data and Write it Up

It's critically important to make detailed notes about any changes you have made when cleaning the data.

```
filenamepath <- glue::glue("./data/{lubridate::today()}-eurovision.rds")
filenamepath
```

```
## ./data/2025-02-18-eurovision.rds
```

```
write_rds(songs,filenamepath)
```